

ISYE 3044

Final Project

Rosheeta Sheth

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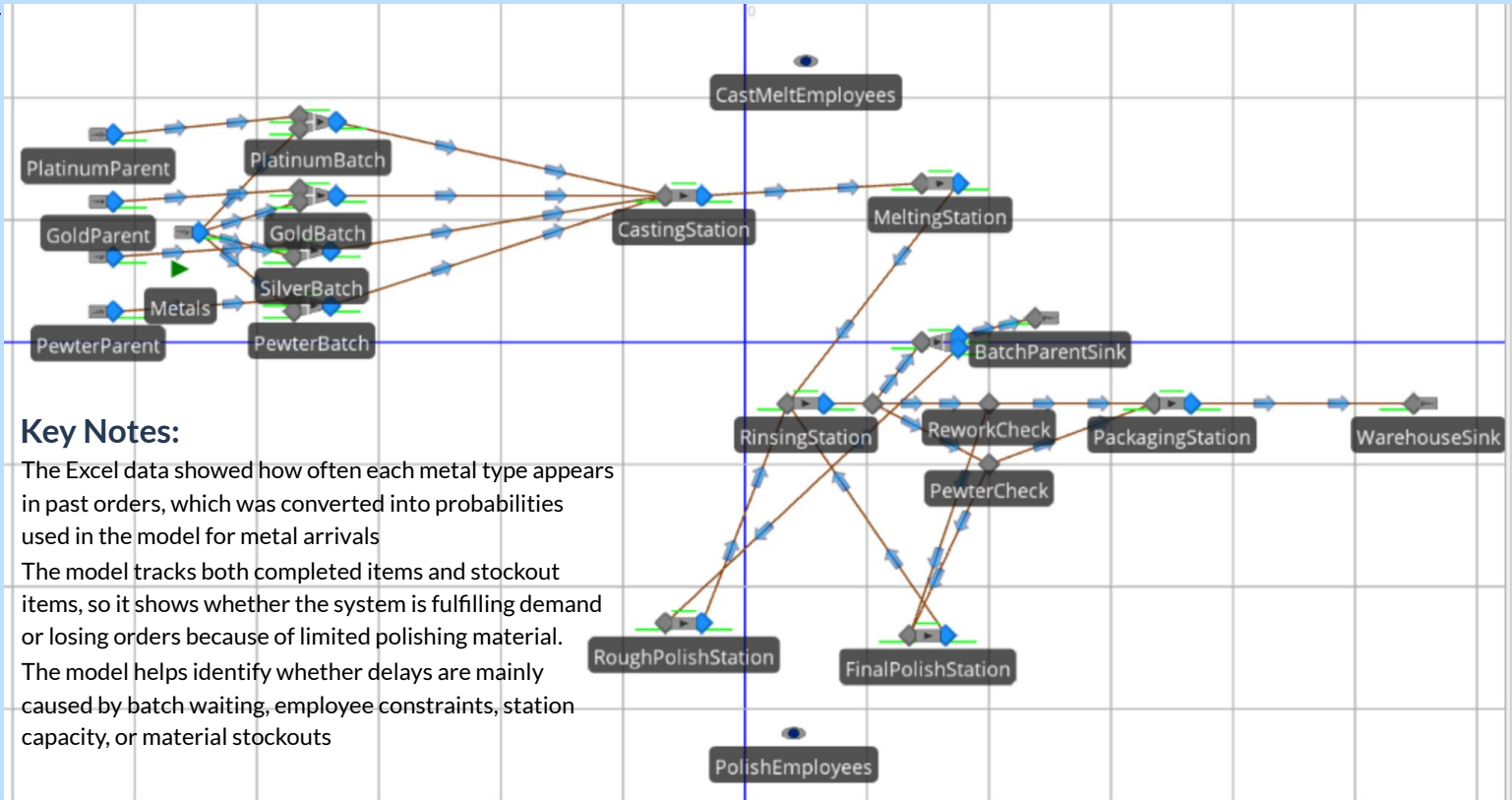
Initial Warehouse System: Model

Process Explanation:

- Orders arrive every Exp(3) minutes and are routed by metal type into PlatinumBatch, GoldBatch, SilverBatch, or PewterBatch.
- Each metal has a different batch size, so the batch combiners hold orders until enough matching items arrive: Platinum uses 4 items, Gold uses 3 items, Silver uses 3 items, and Pewter uses 6 items.
- Completed batches move through CastingStation, MeltingStation, and the shared RinsingStation.
- After the first rinse, BatchSeparator splits the batch back into individual jewelry items, while the unused batch parent is sent to BatchParentSink.
- Individual items go to RoughPolishStation, where polishing material is checked and depleted by metal type.
- Since the model uses one shared RinsingStation, a RinseStage variable is used to send items to the correct next process after rinsing.
- RinseStage is used to route items to BatchSeparator after the first rinse, PewterCheck after rough polish, and ReworkCheck after final polish.
- PewterCheck sends Pewter items around FinalPolishStation, while Platinum, Gold, and Silver continue to FinalPolishStation.
- ReworkCheck handles the 10% chance that Platinum, Gold, and Silver items need one extra Final Polish and rinse before Packaging.
- PackagingStation sends completed items to WarehouseSink, which represents finished orders.
- The model uses CastMeltEmployees for CastingStation and MeltingStation, PolishEmployees for RoughPolishStation and FinalPolishStation, and either employee group for PackagingStation.

Initial Warehouse System: Model

Process Image:



Initial Warehouse System: Results

Model Data:

- Average time in system: 115 minutes, for completed orders that exited through WarehouseSink.
- Number of orders entered into the system: 211 orders.
- Number of orders completed and exited the system: 204 orders exited through WarehouseSink.
- CastMeltEmployees average utilization: 73.58%.
- PolishEmployees average utilization: 73.10%.
- Stockout items:
 - Platinum stockouts: 32, Silver stockouts: 18, Total stockouts: 50 items
 - the estimated stockout cost is \$182:

Issues:

- The current system is losing orders due to polishing material stockouts, especially for Platinum and Silver
- Employee utilization is high for both worker groups, limiting how quickly items move through production
 - Because employees are not cross-trained in the original system, Casting/Melting employees cannot help at polishing stations when polishing demand is high
- Batch waiting can delay orders because each metal type must collect enough matching orders before production can begin

Key Takeaways:

- Most stockouts came from Platinum and Silver, so increasing polishing inventory should directly help reduce lost orders.
- Cross-training all 8 employees could improve labor flexibility and reduce waiting at employee-constrained stations.
- The strongest improvements should focus on reducing stockouts while also improving employee availability across the full process.

Initial Warehouse System: Assumptions

- All processing time distributions are assumed to accurately represent the real jewelry production process.
- Travel time between stations is included based on the Simio layout, but the physical layout is not changed in the initial model.
- Each station processes items using the same rules for every shift, so machine behavior and employee availability are assumed to be consistent across replications.
- Stockout items are considered unfinished orders and are removed from the production system once they enter StockoutSink.
- Reworked items are assumed to require the same FinalPolishStation processing time and rinse time as a regular final polish item.
- The model uses the last 3 digits of the GTID as the random seed so the simulation results can be replicated.

Results Validation — Smell Test

✓ Batch counts correct?

Platinum $31 \times 4 = 124$, Gold $36 \times 3 = 108$, Silver $40 \times 3 = 120$,
Pewter $18 \times 6 = 108 \rightarrow 460$ items total ✓

✓ Flow time reasonable?

Avg 2.02 hrs per entity within a 12-hr shift with multi-step processing ✓

✓ Utilization levels sensible?

All stations between 73-89% – expected for a busy production system ✓

✓ Rinsing station high?

87.2% expected – each entity visits 3 times (post-melt, post-rough, post-final) ✓

✓ Stockout cost plausible?

\$50 total is within daily budget of \$325 – some depletion expected ✓

✓ Not all entities complete?

594 created, 437 reach warehouse – reasonable given 12hr limit & batching wait ✓

Improvement 1 Model Implementation — Cross-Train All 8 Employees

- The change is to cross-train all 8 employees so that any employee can work at CastingStation, MeltingStation, RoughPolishStation, FinalPolishStation, and PackagingStation
- The initial model showed employee constraints and queue buildup, especially around polishing-related steps. Cross-training should help balance labor usage because employees are no longer restricted to only Casting/Melting or only Polishing tasks.
- In Simio, this improvement was implemented by allowing both CastMeltEmployees and PolishEmployees to be secondary resources at the labor-required stations. CastingStation and MeltingStation were updated so they could use either employee group instead of only CastMeltEmployees. RoughPolishStation and FinalPolishStation were updated so they could use either employee group instead of only PolishEmployees. PackagingStation already allowed either employee group, so no major change was needed there.
- This improvement is justified because it increases labor flexibility without adding new employees or equipment. Since training is mainly a one-time cost, it is a reasonable improvement to test before considering more expensive recurring changes like hiring additional workers.

Improvement 1: Results - DIDN'T IMPROVE

Model Data:

- Average time in system: 107.957 minutes, for completed orders that exited through WarehouseSink.
- Number of orders entered into the system: 206 orders.
- Number of orders completed and exited the system: 201 orders exited through WarehouseSink.
- AllEmployee average utilization: 73.18%.
- CastMeltEmployees average utilization: N/A because employees were combined into AllEmployee.
- PolishEmployees average utilization: N/A because employees were combined into AllEmployee.
- Stockout items:
 - Platinum stockouts: 136
 - Total stockouts: 136 items
- The estimated stockout cost is: \$544.

Key Takeaways:

- Increasing polishing inventory improved throughput because more orders were completed. The change directly addressed the stockout issue, but the inventory mix may still need adjustment. This improvement is stronger than cross-training because it increased the number of completed orders

Model after Change

- Compared to the initial model, increasing polishing inventory raised completed orders from 204 to 225.
- This change helped more orders reach WarehouseSink, showing that polishing material availability was limiting the original system.
- Average time in system increased from 114.666 minutes to 120.890 minutes, likely because more orders stayed in the system instead of being removed early through StockoutSink.
- Silver stockouts were eliminated, but stockouts still occurred for Platinum, Gold, and Pewter.

Problems That May Have Occurred:

- Some inventory was reallocated away from Gold and Pewter, which created new stockouts for those metals.
- Platinum still had stockouts, meaning its polishing inventory may still be too low.
- Since more items stayed in production instead of leaving as stockouts, queues and cycle time increased.

Improvement 1: Assumptions

- All 8 employees can be cross-trained to work at CastingStation, MeltingStation, RoughPolishStation, FinalPolishStation, and PackagingStation.
- It is assumed that cross-training does not change the processing times at each station, and only changes which employees are allowed to work at each station.
- Cross-training is mainly a one-time training cost and does not require hiring additional employees or purchasing new equipment.

Improvement 2 Model Implementation — Increase Polishing Inventory

- The change is to increase the starting polishing material inventory for the metals with the highest stockouts
- The initial model showed that stockouts were causing some orders to leave the system before completion. Platinum and Silver had the largest stockout issue, so the improvement focuses on increasing their available polishing material.
- In Simio, this improvement was implemented by changing the beginning inventory values for polishing material before running the experiment. The stockout logic stayed the same: if an item did not have enough polishing material available, it was still sent to StockoutSink. RoughPolishStation and FinalPolishStation still depleted the correct metal-specific inventory whenever an item was polished.
- This improvement is justified because it directly targets the reason orders were being lost in the initial system. Increasing inventory is a recurring cost, but it may be worthwhile if it reduces stockouts and allows more orders to reach WarehouseSink.

Model Data:

- Average time in system: 120.890 minutes, for completed orders that exited through WarehouseSink.
- Number of orders entered into the system: 229 orders.
- Number of orders completed and exited the system: 225 orders exited through WarehouseSink.
- CastMeltEmployees average utilization: 79.56%.
- PolishEmployees average utilization: 79.96%.
- Stockout items:
 - Platinum stockouts: 64
 - Silver stockouts: 0
 - Gold stockouts: 28
 - Pewter stockouts: 24
 - Total stockouts: 116 items
- The estimated stockout cost is: \$404.

Key Takeaways:

- Since employees were more flexible, items may have moved through labor-required stations faster and reached polishing material limits sooner.
- The model still had the same starting polishing inventory, so inventory shortages remained.

Model after Change

- Compared to the initial model, cross-training all 8 employees slightly reduced the average time in system from 114.666 minutes to 107.957 minutes.
- The shared employee pool allowed workers to be used more flexibly across CastingStation, MeltingStation, RoughPolishStation, FinalPolishStation, and PackagingStation.
- However, completed orders slightly decreased from 204 to 201, and stockouts increased from 50 to 136.
- This suggests that labor flexibility helped reduce some waiting time, but it did not fix the polishing material inventory issue.

Problems That May Have Occurred:

- Since employees were more flexible, items may have moved through labor-required stations faster and reached polishing material limits sooner.
- The model still had the same starting polishing inventory, so inventory shortages remained.
- Cross-training improves employee constraints, but it does not directly increase the number of orders that can be completed if material runs out.
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Improvement 2: Assumptions

- Management can adjust the starting polishing material inventory while staying close to the \$325 daily budget.
- It is assumed that the added or reallocated polishing material is available at the beginning of the shift.
- Changing inventory does not change processing times, employee requirements, station capacities, or the production layout.

Final Recommendations

Improvement 2: Increase Polishing Inventory

- This change is recommended because the initial model showed that stockouts were one of the main reasons orders were not being completed. Increasing polishing inventory directly addresses this issue by allowing more items to continue through RoughPolishStation and FinalPolishStation instead of being sent to StockoutSink.
- This improvement is more directly connected to the client's goal of fulfilling all orders while keeping costs low. Cross-training improves labor flexibility, but it does not solve the issue of orders being lost when polishing material runs out.
- The inventory increase is especially justified because most stockouts came from Platinum and Silver. By increasing the starting inventory for the materials with the highest stockout counts, the system can reduce lost orders without changing the production layout or adding new equipment.
- Although purchasing additional polishing material is a recurring cost, it can be justified if the reduction in stockout cost and lost orders is greater than the added inventory cost.

General Observations Made

- The main issue in the initial system was not only processing speed, but also whether enough polishing material was available to complete incoming orders.
- Cross-training all 8 employees may help reduce waiting caused by employee availability, but it does not prevent stockouts if material inventory is too low.
- Increasing inventory should mainly improve the number of completed orders and reduce the number of items sent to StockoutSink.
- Remaining delays may still occur because orders must wait to form batches before entering CastingStation.
- Some bottlenecks may still remain at polishing or packaging if demand is high, especially when Platinum, Gold, or Silver items require rework.
- If management wants further improvement after reducing stockouts, the next area to study would be labor flexibility through cross-training or additional capacity at high-queue stations.

Thank You!